COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010	SECTION: CPE21S4	
DATE PERFORMED:	13/08/2026	DATE SUBMITTED:	13/08/2026
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.

Supplementary:

main.cpp stack.cpp

supplementary > main.cpp > ...

```cpp
#include <iostream>
using namespace std;

#define MAX 100

class Stack {
    char arr[MAX];
    int top;
public:
    Stack() { top = -1; }

    void push(char c) {
        if (top == MAX - 1) {
            cout << "Stack Overflow\n";
            return;
        }
        arr[++top] = c;
    }

    char pop() {
        if (top == -1) {
            cout << "Stack Underflow\n";
            return '\0';
        }
        return arr[top--];
    }

    bool isEmpty() { return top == -1; }
};

bool isBalanced(const string &expr) {
    Stack s;
    for (char c : expr) {
        if (c == '(' || c == '{' || c == '[')
            s.push(c);
        else if (c == ')' || c == '}' || c == ']') {
            if (s.isEmpty()) return false;
            char top = s.pop();
            if ((c == ')' && top != '(') ||
                (c == '}' && top != '{') ||
                (c == ']' && top != '['))
                return false;
        }
    }
    return s.isEmpty();
}

int main() {
```

```cpp
int main() {
    string expressions[] = {"(A+B)+(C-D)", "((A+B)+(C-D)", "(A+B)+(C-D))", "((A+B)+(C-D))"};
    for (string expr : expressions) {
        cout << expr << " -> " << (isBalanced(expr) ? "Valid" : "Invalid") << endl;
    }
    return 0;
}
```

main.cpp | stack.cpp

supplementary > stack.cpp > ...

```cpp
#include <iostream>
using namespace std;

class Node {
public:
    char data;
    Node* next;
    Node(char c) { data = c; next = nullptr; }
};

class Stack {
    Node* top;
public:
    Stack() { top = nullptr; }

    void push(char c) {
        Node* newNode = new Node(c);
        newNode->next = top;
        top = newNode;
    }

    char pop() {
        if (!top) {
            cout << "Stack Underflow\n";
            return '\0';
        }
        char c = top->data;
        Node* temp = top;
        top = top->next;
        delete temp;
        return c;
    }

    bool isEmpty() { return top == nullptr; }
};

bool isBalanced(const string &expr) {
    Stack s;
    for (char c : expr) {
        if (c == '(' || c == '{' || c == '[')
            s.push(c);
        else if (c == ')' || c == '}' || c == ']') {
            if (s.isEmpty()) return false;
            char top = s.pop();
            if ((c == ')' && top != '(') ||
                (c == '}' && top != '{') ||
                (c == ']' && top != '['))
```

```
                return false;
            }
        }
        return s.isEmpty();
    }

    int main() {
        string expressions[] = {"(A+B)+(C-D)", "((A+B)+(C-D)", "(A+B)+(C-D))", "((A+B)+(C-D))"};
        for (string expr : expressions) {
            cout << expr << " -> " << (isBalanced(expr) ? "Valid" : "Invalid") << endl;
        }
        return 0;
    }
```

```
[Running] cd "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\supplementary\" && g++ main.cpp -o main && "c:\Users\Jullian Sevilla\OneDrive\Documents\DSA\supplementary\"main
(A+B)+(C-D) -> Valid
((A+B)+(C-D) -> Invalid
(A+B)+(C-D)) -> Invalid
((A+B)+(C-D)) -> Valid

[Done] exited with code=0 in 1.895 seconds
```